

CMF Mini Game Console – Hardware Documentation

This document lists the hardware components, buying references, and pin connections for the CMF Mini Game Console project.

1. Components Used

Component	Reference / Buying Page
ESP32-C3 Development Board	robu.in/product/esp32-c3-development-board-with-soldering
1.3 inch OLED Display (SH1106)	robu.in/product/1-3-oled-display-v2
Analog 5-Key Button Module	robu.in/product/electronic-building-emulation-keyboard-module

2. Pin Connections

OLED Display (SH1106 – I2C)

OLED Pin	ESP32-C3 Pin
VCC	3.3V
GND	GND
SDA	GPIO 8
SCL	GPIO 9

Analog Button Module

Module Pin	ESP32-C3 Pin
VCC	3.3V
GND	GND
SIG	GPIO 0 (ADC)

3. Schematic Overview

The ESP32-C3 communicates with the OLED display using the I2C interface. The analog keypad uses a resistor ladder and connects to a single ADC pin. Power is supplied through the USB-C port or external 5V source.